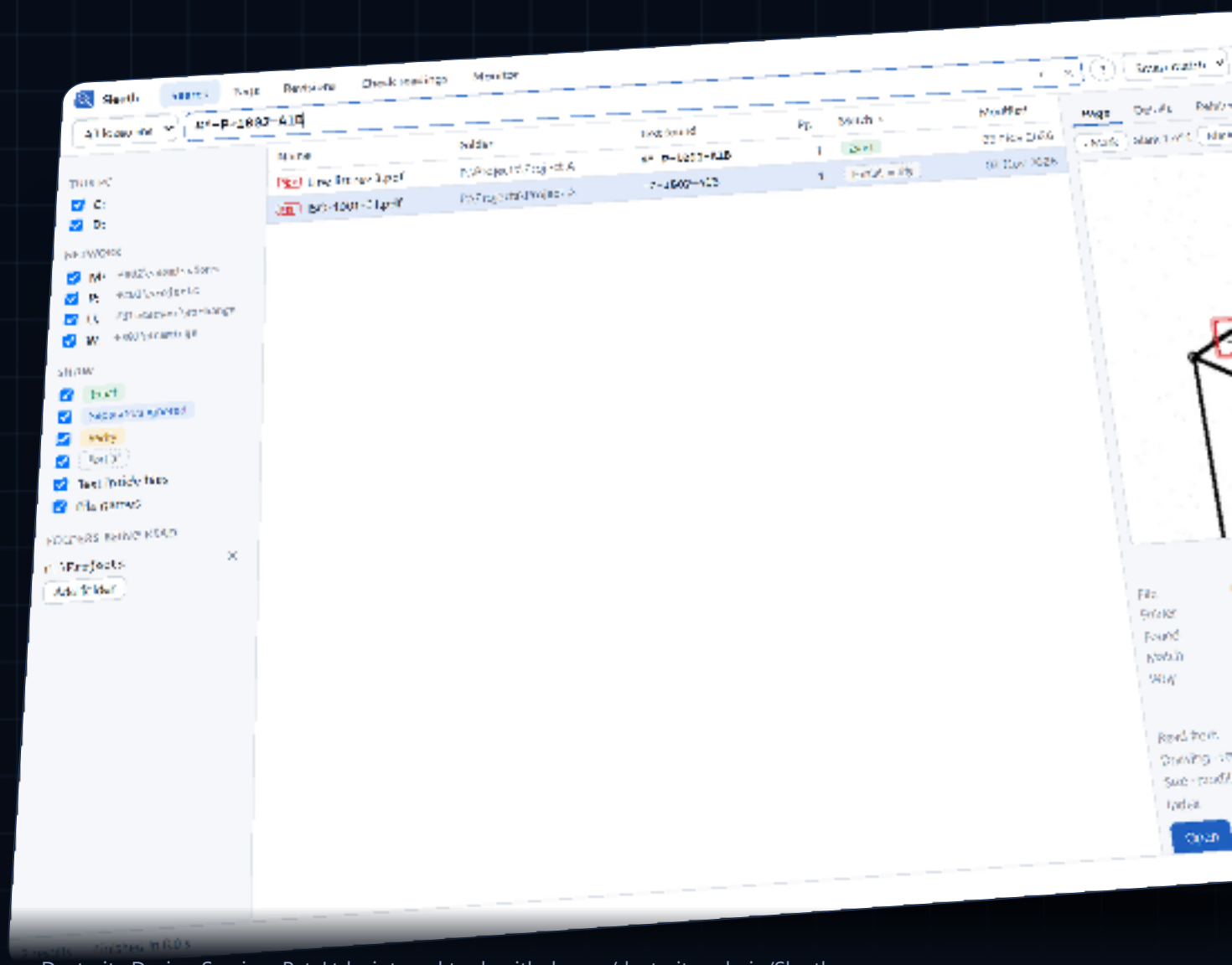
USER GUIDE · VERSION 1.0

Sleuth.

Find any tag, in any drawing, on any drive. Even when the PDF is a scan.



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Find by any text
Tag numbers, line numbers, document numbers, notes:

Even in scans
Sleuth reads scanned and plotted PDFs with OCR, so text

See the proof
Each result opens on its page with the text circled in red.

anything printed on a drawing.

with no text layer is found too.

1 What Sleuth is

Sleuth is our office's search tool for drawings and documents. Type a tag and it shows every file that contains it, which page it's on, and the spot on the page.

Windows search, and tools like Everything, only look at file names, or at text that a PDF already carries. Most of our drawings are scans or plots with no text inside, so those tools can't find a line number printed on an isometric. Sleuth reads every page once, including scans, and remembers what it read. Searches then take a fraction of a second.

What Sleuth does for you

- **Finds text inside drawings**, including scanned and plotted PDFs, with OCR that reads small print, rotated and vertical text.
- **Understands tags**: `6"-P-1001-A1B`, `6-P-1001-A1B` and `6P1001A1B` are the same tag to Sleuth.
- **Searches every drive**: your own disks and every mapped network drive, without freezing when a share is slow or offline.
- **Shows the evidence**: every hit opens on its page with the text marked, so you confirm it with your own eyes.
- **Shares the work**: a drawing read on any PC in the office is not read again on yours.

How it works, in one paragraph

Sleuth has two parts. The **Sleuth window** is what you search in. The **Sleuth reader** is a background program that keeps a list of file names on your drives and reads PDFs in the folders you choose. It pauses when you're busy and speeds up when the PC is idle. Everything it learns is kept in an index on your PC, plus the office's shared reading cache, which is why searches are instant.

> **For engineering use**
> Sleuth helps you find drawings; it does not certify them. When a result is labelled `Verify, OCR unsure`, or when the tag matters for a deliverable, open the page and read the tag yourself. Sleuth always shows you where to look.

2 Installing Sleuth

Sleuth installs for your Windows account only. You don't need administrator rights, and it takes about a minute.

1. Open **github.com/dexterity-admin/Sleuth/releases/latest** and download `Sleuth_1.0.0_x64-setup.exe` (about 65 MB), or use the copy IT shared.
2. Run it. Because the installer is not code-signed, Windows may show **"Windows protected your PC"**. Click **More info**, then **Run anyway**. You only see this for the first install; updates install from inside Sleuth.
3. Click through the installer. Sleuth starts when it finishes, and adds a Start-menu entry.

> **For IT**
> Silent install: `Sleuth_x.y.z_x64-setup.exe /S`. To choose the folder, add `/D=C:\Tools\Sleuth` as the last argument.

What gets installed

Item	Where	What it is
Sleuth	`%LOCALAPPDATA%\Programs\Sleuth`	The window (`sleuth.exe`), the reader (`sleuth-worker.exe`), the PDF engine and the OCR models
Your settings	`%APPDATA%\Sleuth\settings.json`	Folders to read, cache folder and options. Updates never touch this file.
Your index	`%LOCALAPPDATA%\Sleuth`	The file-name list, the text read from PDFs, OCR models and logs

Requirements

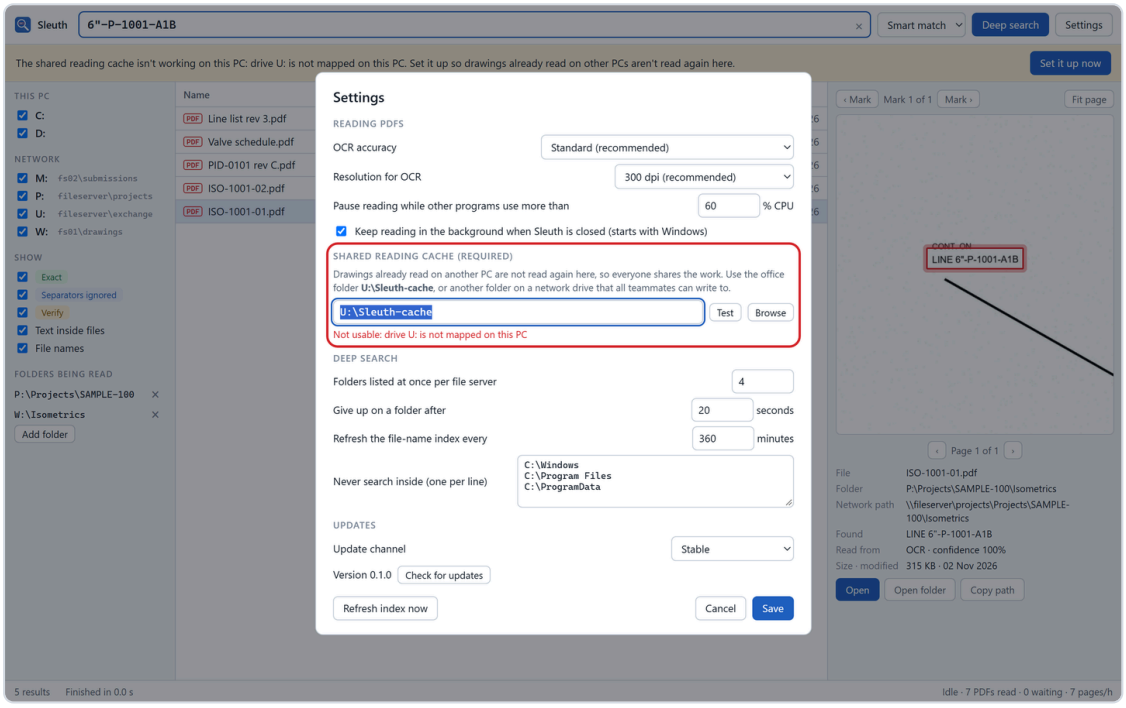
- Windows 10 or 11, 64-bit.
- While it is reading drawings, the reader uses about 1.6 GB of memory and part of the CPU. It frees the memory after 2 minutes with nothing to read, and the window itself uses about 150 MB.
- About 500 MB of disk for the program and a typical index.

3 First-time setup

Two things need to be right: the shared reading cache, and the folders you want Sleuth to read. Both take a minute.

Step 1: the shared reading cache

Sleuth uses the office folder `U:\Sleuth-cache` automatically. If your PC has `U:` mapped, there is nothing to do. If it doesn't, Sleuth shows a yellow message at the top of the window. If you leave it, Sleuth opens Settings and highlights the place to fix it:



Sleuth won't let this go: until the cache works, it asks every minute. The section on the shared reading cache explains why.

1. Type the folder IT gave you, or click **Browse** to pick it. It must be on a network drive everyone can write to.
2. Click **Test**. "Working" in green means it's right.
3. Click **Save**. Sleuth refuses to save a folder that doesn't work, and tells you why.

Step 2: choose the folders to read

Sleuth always knows the **names** of files on all your drives. To search **inside** PDFs, it has to read them first, and it only reads the folders you choose. Add the project folders you work in:

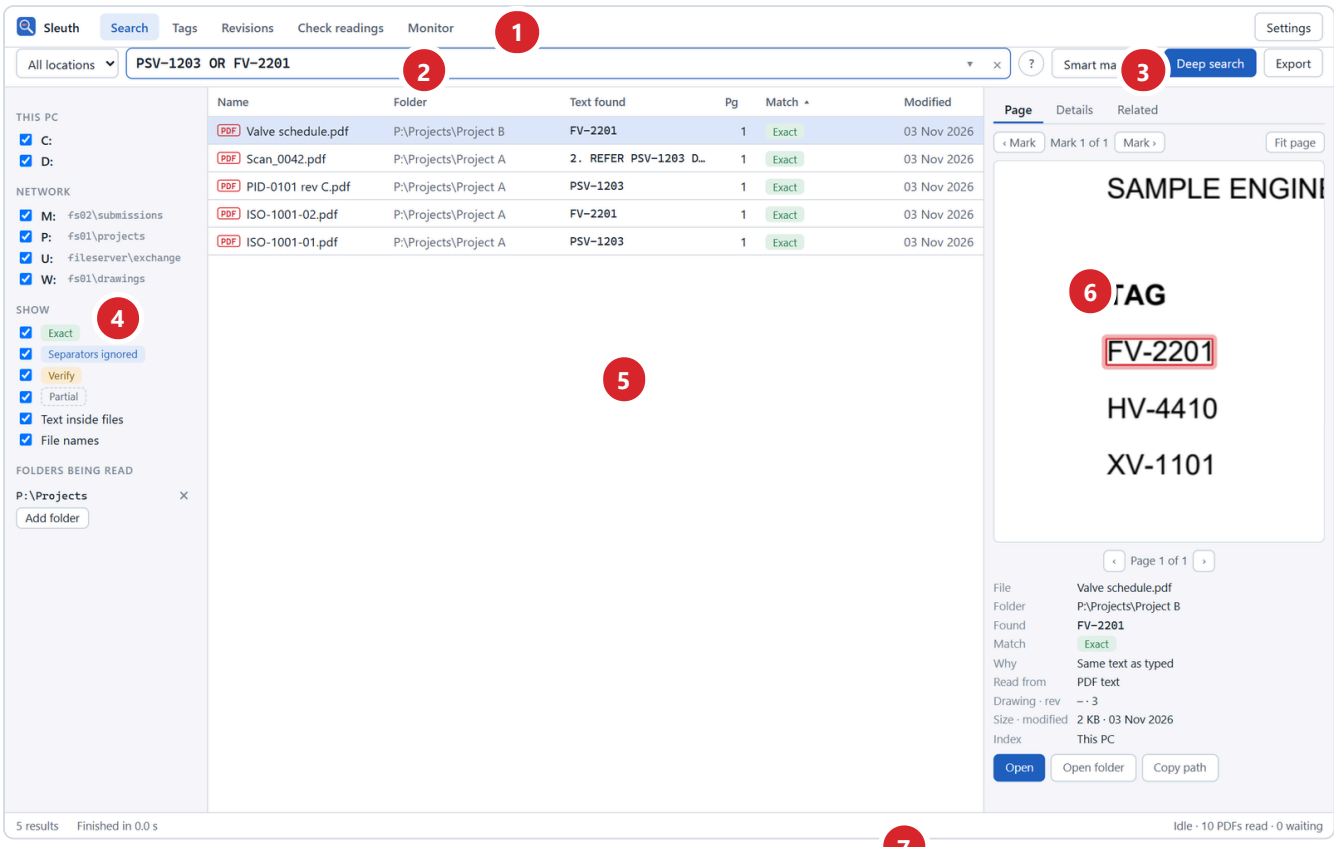
1. In the left pane, under **Folders being read**, click **Add folder**.
2. Pick the folder, for example `P:\Projects\SAMPLE-100`. Sleuth starts reading its PDFs in the background.
3. To stop reading a folder, click the × next to it.

How long does reading take?

PDFs with a text layer are read in seconds. A scanned A1/A3 sheet takes about a minute on an office PC. Pages already read on another PC come from the shared cache instantly. You can search at any time; results improve as reading continues.

4 A tour of the window

Sleuth's window has a search bar at the top and three panes below it.



A search for two tags at once. The numbers match the table below.

#	Part	What it does
1	Views	**Search**, **Tags**, **Revisions**, **Check readings** and **Monitor** (sections 10 to 14).
2	Project and search box	Pick a project, or **All locations**. Type a tag or any text; results appear as you type. `Ctrl` + `F` jumps here. **?** shows the search syntax; ▾ shows recent and pinned searches.
3	Smart match, Deep search, Export	**Smart match** also finds tags written differently; **Exact only** hides those. **Deep search** walks the drives live (section 8). **Export** saves the results to Excel or CSV.
4	Locations and filters	Tick the drives to search. Show or hide result types. Manage **Folders being read**.
5	Results	One row per file: name, folder, the exact text found, page and match type. Hover a row to see why it matched. Click a column heading to sort.
6	Preview	**Page**: the page with the text circled in red. **Details**: title block and how the file was read. **Related**: drawings sharing tags.
7	Status bar	Number of results, the office server if you use one, and what the reader is doing. Click the reader status for details.

5 Searching

Type what you're looking for. Plain text always works; a few extras narrow things down.

What you can search for

Looking for	Type	Notes
A line number	`6"-P-1001-A1B`	Quotes, dashes and spaces are optional: `6P1001A1B` works too.
An instrument or valve tag	`PSV-1203` , `FV 2201`	A tag typed with a space is still found.
A document number	`DOC-1001-17`	Finds the document itself and every drawing that refers to it.
A drawing number or file name	`ISO-1001-02`	File names are searched on every drive, read or not.
Several things at once	`PSV-1203 flare`	Files that contain **all** the words, anywhere in the file.
Any of several tags	`PSV-1203 OR PSV-1204`	`OR` (capitals) or `|` between alternatives.
Drawings by title block	`dwg:ISO-1001 rev:C`	Drawing number contains, revision. Also `disc:piping` .
Only one kind of tag	`type:instrument FV`	Kinds: line, instrument, valve, equipment, support, weld, drawing, spec.
Words in the file name	`name:datasheet`	`ext:pdf` limits the file type.
An exact phrase	`"rev c"`	Put the phrase in double quotes. An inch mark inside a tag, as in `6"-P` , is never treated as a quote.

Narrowing the results

- **Project**: pick a project next to the search box to search only its folders (section on projects).
- **Drives**: untick a drive in the left pane to leave it out.
- **History**: the ▾ in the search box lists recent searches; pin the ones you use often.
- **Export**: the results to Excel or CSV, with the reason for each match.
- **Show**: hide Exact , Separators ignored or Verify results, or results found only in file names.
- **Exact only**: the drop-down next to the search box hides everything except exact matches.
- **Sort**: click a column heading; click again to reverse.

Keyboard

Key	Does
`Ctrl` + `F`	Go to the search box
`Enter` in the search box	Search now; `Ctrl` + `Enter` runs a deep search
`Esc`	Clear the search
`↓` / `↑`, `Page Down` / `Page Up`	Move through results; the preview follows
`Enter` on a result, or double-click	Open the file

6 How results are labelled, and when to double-check

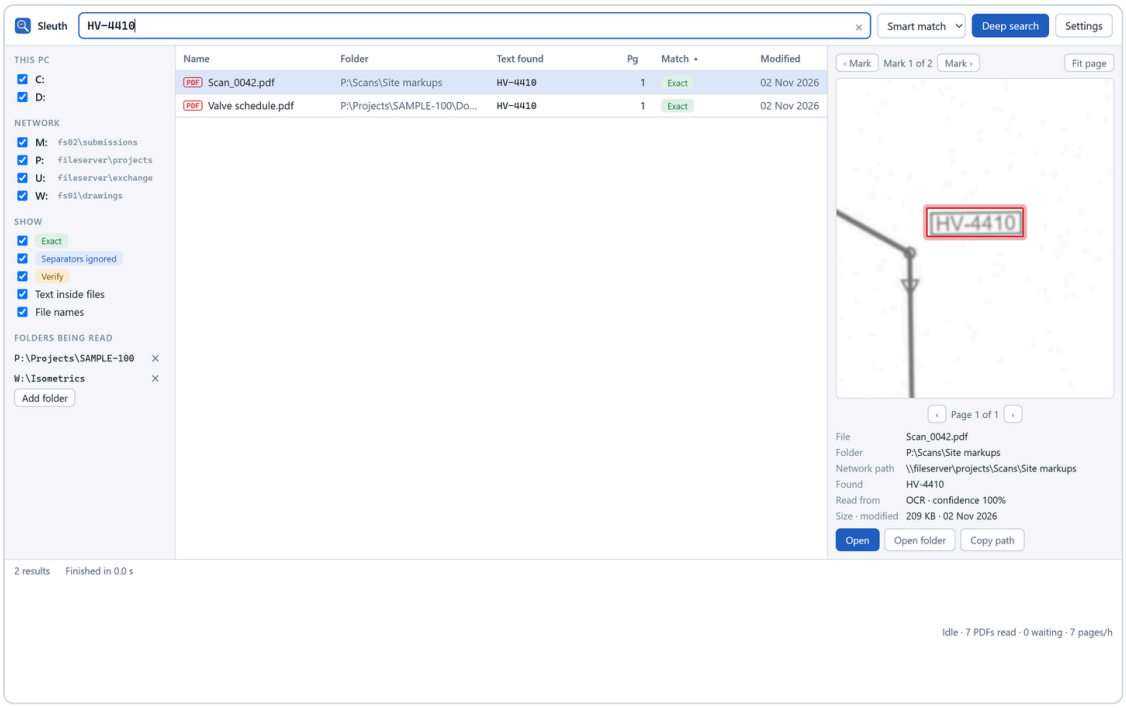
Every result says how it matched and where the text came from. This is what lets you trust Sleuth on engineering work.

Label	Means	What to do
Exact	The text on the page is exactly what you typed.	Nothing; open it.
Separators ignored	Same letters and digits, written with different dashes, quotes or spaces. `PSV1203` found `PSV-1203`.	Usually right. Glance at the preview.
Verify, OCR unsure	Either the OCR wasn't confident, or a character could be a look-alike (O/0, I/1, S/5, B/8, Z/2, G/6).	**Check the preview** before relying on it.
Partial, verify	Only part matched: your text is inside a longer tag (`PSV-120` in `PSV-1203`), or only part of the tag could be read, for example because a pipe line crosses the size (`-P-1002-A1B` for `4"-P-1002-A1B`).	**Look at the page.** The row's "Why" says which.
Name · ...	Found in the file name, not inside the file.	Useful when the PDF hasn't been read.

Where the text came from

Hover over a result to see why it matched and where the text came from; the preview pane shows both.

- **PDF text**: the PDF carries real text (exported from CAD or Word). Always exact.
- **OCR**: Sleuth read the image of the page, with a confidence percentage. When the main reader was unsure, Windows' own OCR read the line again. If the two disagree, both readings are kept and the result is marked Verify.
- **Checked**: a person typed this reading in Check readings.
- **File name**: matched the name only.



A low-contrast site scan. The tag was read with 100% confidence; the circled mark lets you confirm it in a second.

Rule of thumb

Sleuth never reports an uncertain reading as Exact. If it matters, look at the page: the preview is one click away and always shows the spot.

7 Previewing and opening files

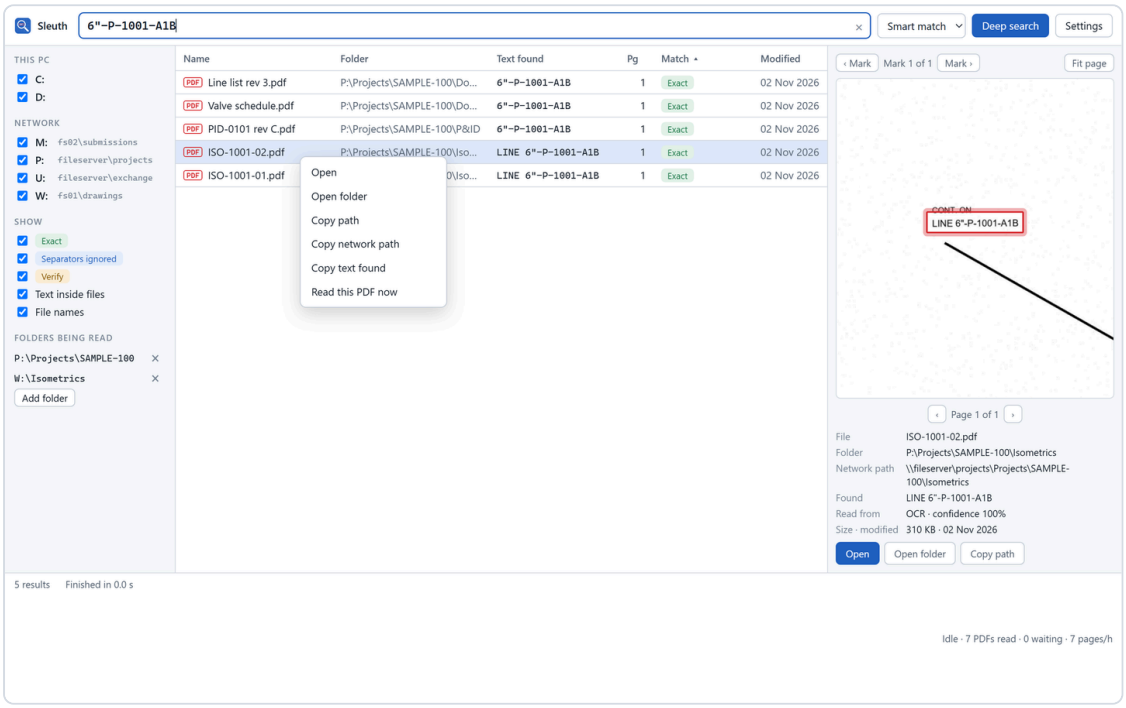
Click a result and the page appears on the right, with the text circled.

In the preview pane

- ‹ **Mark / Mark** › steps between every place the text appears on the page.
- **Zoom to mark / Fit page** switches between a close-up and the whole sheet.
- ‹ › under the page moves between pages.
- The details list the file, its folder, its network path, what was found, how it was read, and its size and date.
- **Open**, **Open folder** and **Copy path** are under the details.

Right-click a result

- **Open** or **Open folder** (the file is selected in Explorer).
- **Copy path**, with your drive letter.
- **Copy network path**: the `\\server\share` form, which works on any PC whatever drive letters it uses.
- **Copy text found**: the exact text read.
- **Read this PDF now**: moves the file to the front of the reading queue.



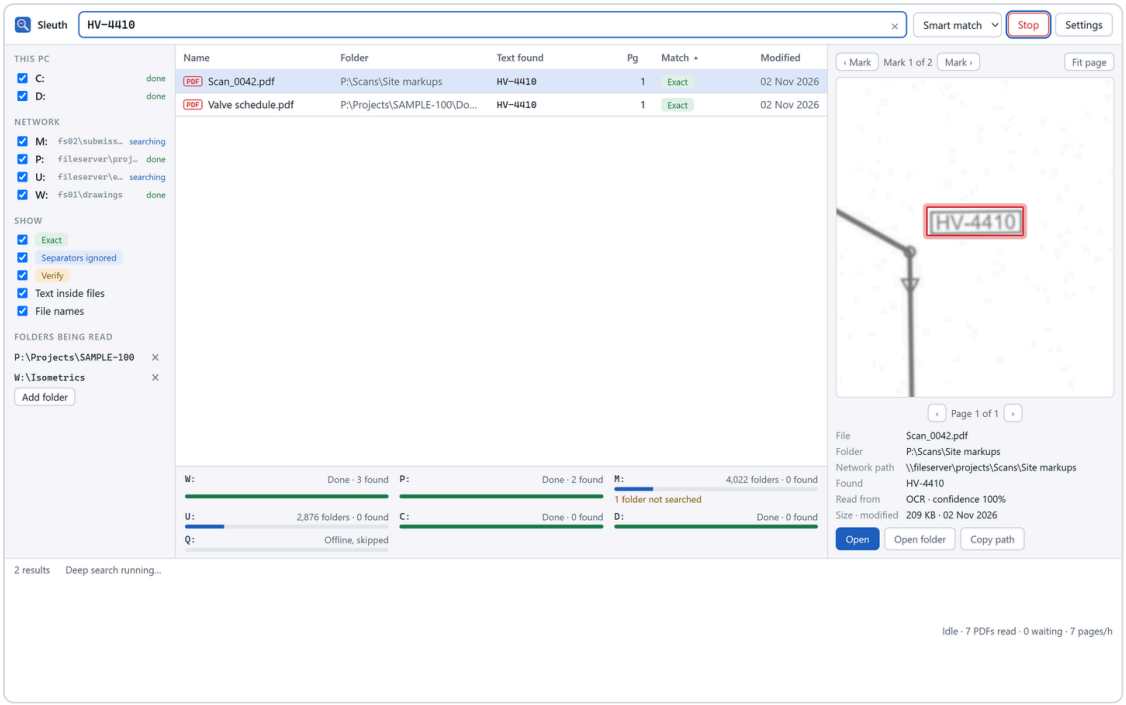
Right-click any result for these actions.

> **Sharing a location with a colleague**
> Use **Copy network path**. Drive letters differ between PCs; the network path doesn't.

8 Deep search: when you don't know where it is

A normal search looks in what Sleuth already knows. A deep search also walks the drives live, right now.

Use it when a file is new, sits outside the folders being read, or may be on a drive Sleuth hasn't listed yet. Type your search and click **Deep search** (or press `Ctrl` + `Enter`).



One bar per drive. Finished drives turn green; an unreachable share is skipped in seconds.

- **Progress per drive** shows under the results: folders searched and hits found. The left pane shows each drive's state.
- **Unread PDFs found on the way** are read straight away, ahead of everything else. They join the results as soon as they're read, so leave Sleuth open for a minute after the walk finishes.
- **Offline drives** are skipped after a few seconds instead of hanging the search.
- **Slow or locked folders** are given up on after the time set in Settings, and listed as "folders not searched". Hover over that line to see which ones and why.
- Drives on the same file server share a speed limit, so the server isn't overloaded.
- Click **Stop** at any time. Results found so far stay.

> **Search fewer drives for speed**
> Untick drives you know aren't relevant in the left pane before starting a deep search.

9 Background reading and the index

The Sleuth reader works quietly in the background so searches are instant.

What the reader does

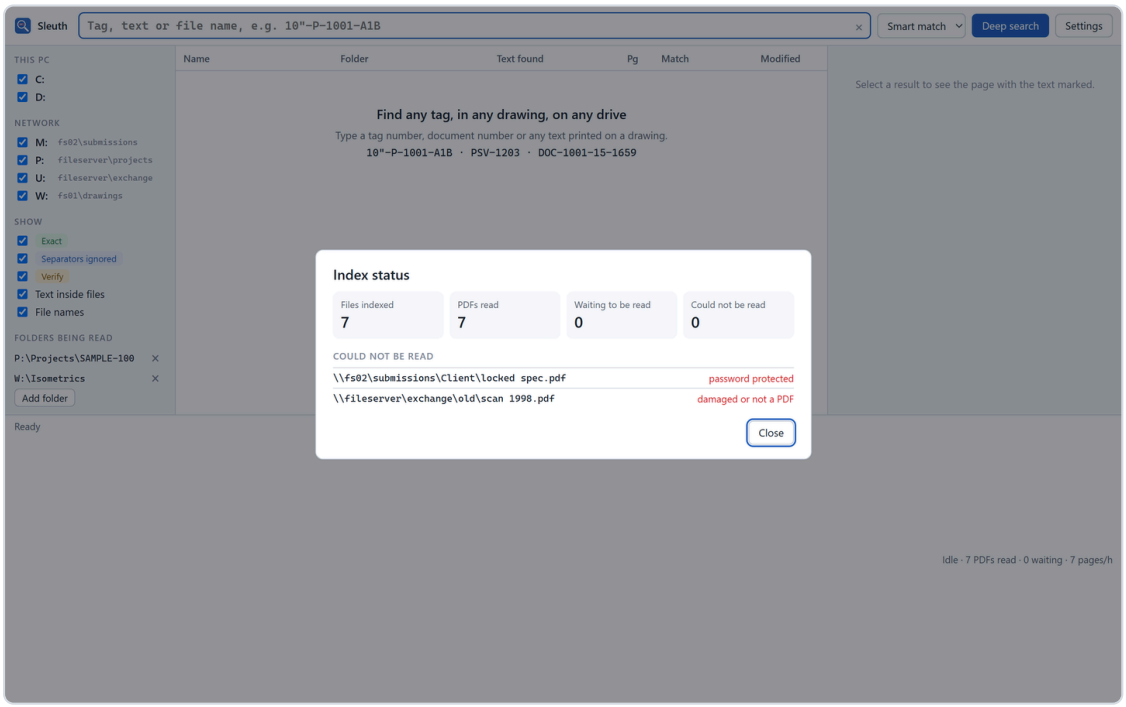
- **File names**: lists the files on every local and network drive, and refreshes the list every 6 hours.
- **Reading PDFs**: reads the PDFs in your **Folders being read**, newest first. Files you ask for (deep search, "Read this PDF now") always go first.
- **Changes**: a file that changes is read again automatically.

Being polite to your PC

- Reading pauses while other programs keep the CPU busy (above 60% by default) and while the PC runs on battery.
- It uses half the CPU cores while you work, and more when nobody has touched the PC for 5 minutes.
- The OCR engine is unloaded from memory after 2 minutes with nothing to read.
- With "Keep reading in the background" on (the default), the reader keeps going after you close the window, and starts with Windows.

Index status

Click the reader status in the bottom-right corner to see how many files are indexed, read and waiting, and which files could not be read and why.



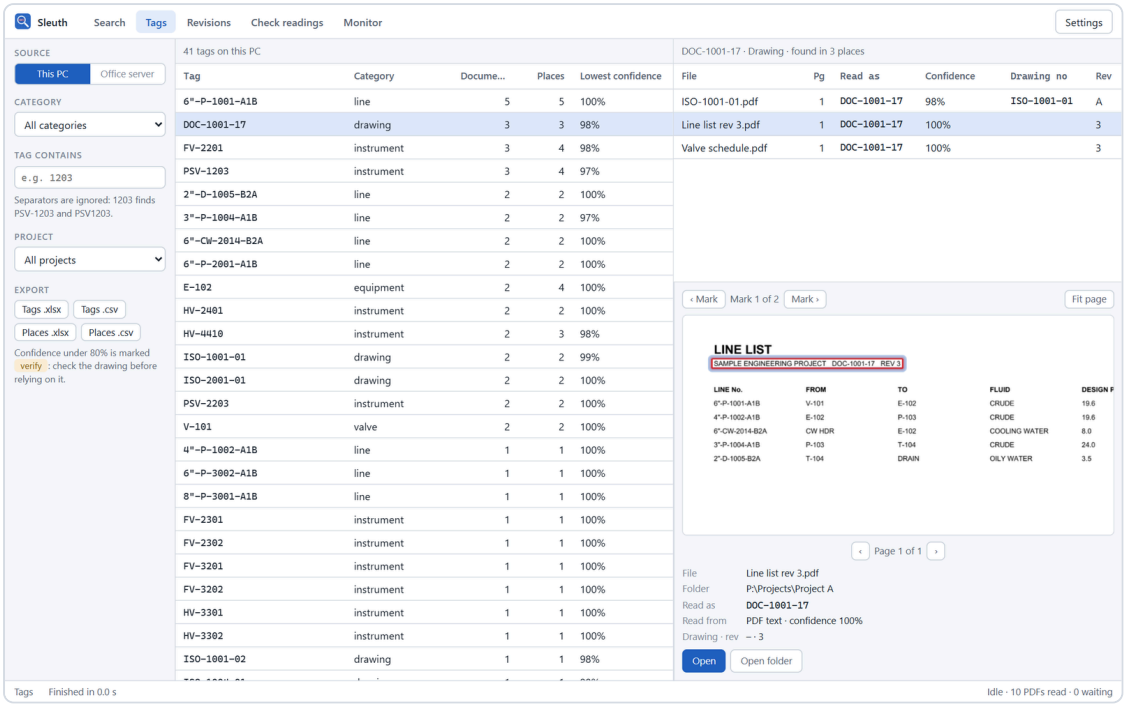
Files that can't be read are listed with the reason, never silently skipped.

Reason shown	Meaning

password protected	The PDF is encrypted. Sleuth can still find it by name.
damaged or not a PDF	The file is corrupt, or has a .pdf name but isn't one.
could not be opened: …	The file moved, is locked, or you have no access. Network hiccups are retried automatically: up to 5 times, at growing intervals, before a file is listed here.

10 Tags: every tag and where it appears

The **Tags** view lists every engineering tag Sleuth has found in the drawings it has read, with the documents and places it appears in.



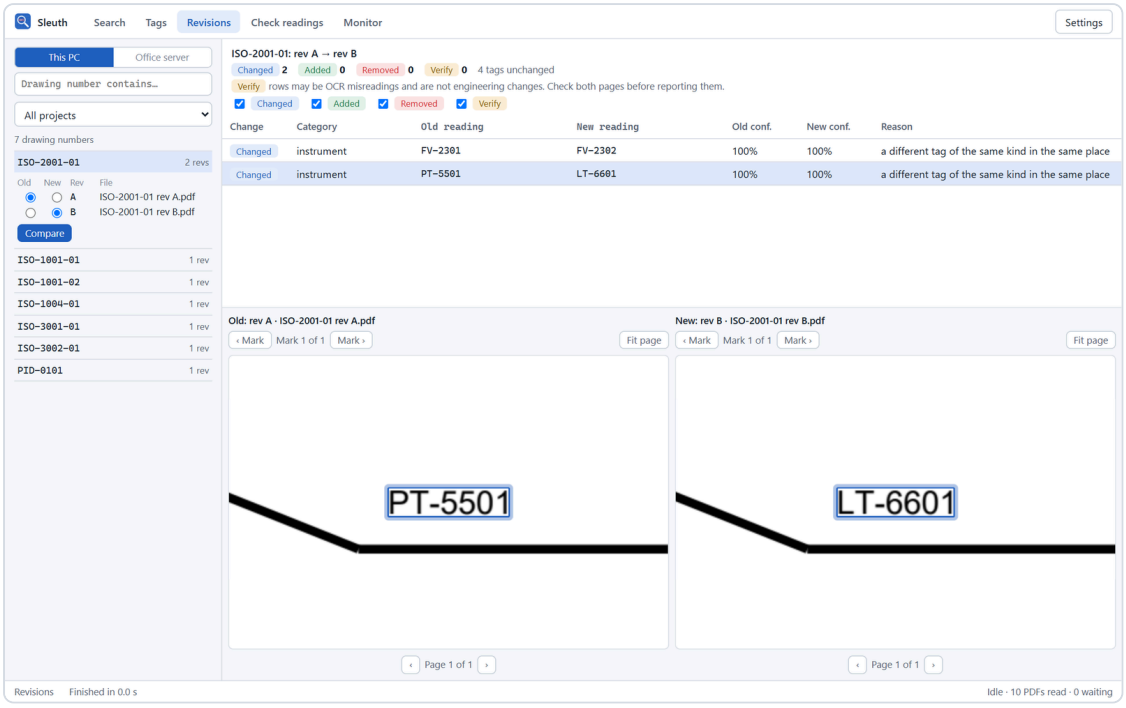
Click a tag to list every place it appears; click a place to see the page with the tag marked.

- **Categories**: line numbers, instruments, valves, equipment, supports, welds, drawing numbers and specs. Sleuth finds them with fixed patterns (rules) applied to the text it read; nothing is guessed. A line number like `6"-P-1001-A1B` is one tag, never also "equipment P-1001".
- **Lowest confidence**: the weakest reading among the places. Under 80% it is marked verify : check those places on the drawing.
- **Filters**: category, "tag contains" (separators are ignored, so `1203` finds `PSV-1203`), project, and **This PC** or **Office server**.
- **Export**: tags or places to Excel (.xlsx) or CSV, for line lists and checks.

> **For the CAD lead**
> If your project numbers tags differently, the rules can be changed in Settings → Advanced (tag rules). Sleuth parses the drawings already read again with the new rules, without reading them again.

11 Revisions: what changed between two issues

The **Revisions** view groups drawings by drawing number and compares the tags of any two revisions.



Rev A against rev B. Select a change to see both sheets with the tag marked.

1. Find the drawing number (type part of it in the filter).
2. Expand it. Sleuth lists its revisions oldest first and picks the last two; choose others with **Old** and **New**.
3. Read the changes:

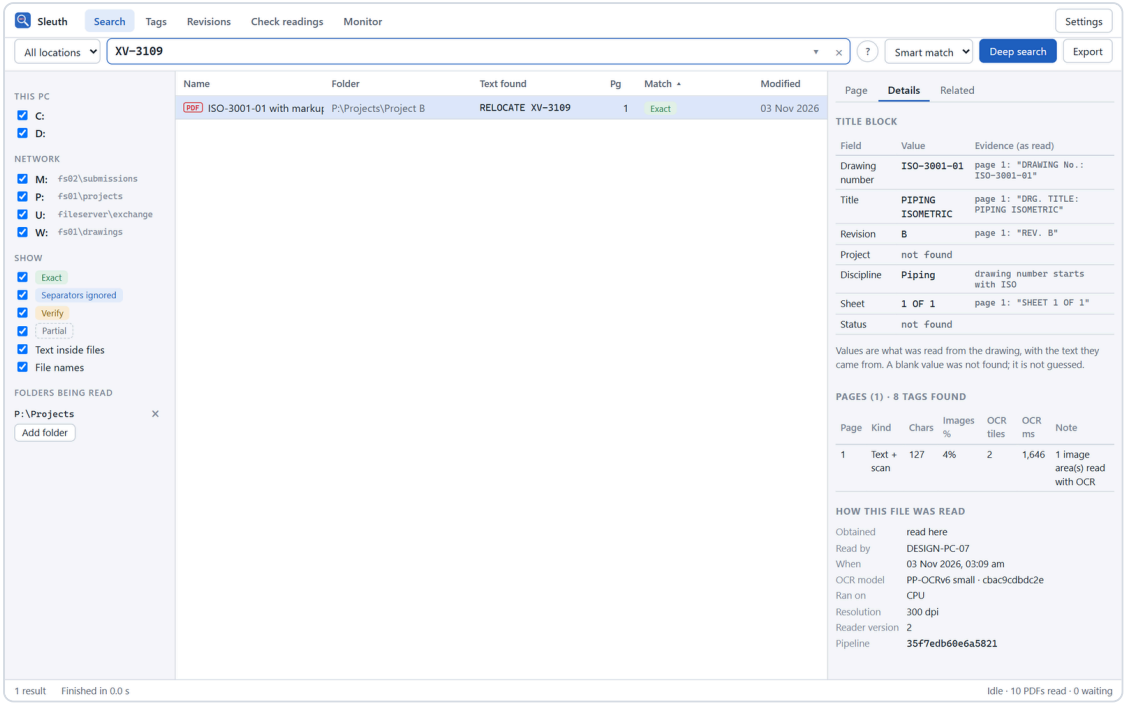
Change	Means
Changed	A different tag of the same kind in the same place on the sheet.
Added / Removed	A tag only in the new, or only in the old revision.
Verify	The difference could be an OCR misreading: the two readings differ only by look-alike characters, or one of them was read with low confidence.

Not an engineering judgement

Sleuth compares what it read. A **Verify** row is never an engineering change until you have looked at both pages. Revision numbers and drawing numbers come from the title block; Details (next section) shows the text each was read from.

12 Details and related drawings

The preview pane has three tabs: **Page**, **Details** and **Related**.



Every value shows the text it was read from. A value Sleuth did not find says "not found"; it is never guessed.

- **Title block**: drawing number, title, revision, project, discipline, sheet and status, each with its evidence (for example `page 1: "REV. B"`, or `OCR 98%` when it was read from a scan).
- **Pages**: what kind each page is and how it was read:

Kind	Means	Read with OCR
Text layer	A plotted or exported PDF with real text	No (only the title block, if its text has no drawing number)
Text + scan	Real text plus pasted images, stamps or scanned mark-ups	Only the images
Scanned	A scan, or a picture of the drawing	The whole page
Outlined text	Text plotted as lines	The whole page
Damaged	The page cannot be opened	No; the reason is shown and the other pages are read

- **How this file was read**: on which PC, when, which OCR model (with its fingerprint), CPU or graphics card, and whether it came from the shared cache.
- **Related**: other drawings that share tags with this one (line numbers, instruments, equipment...), most shared first. Click one to open it on the page where a shared tag appears.

13 Check readings

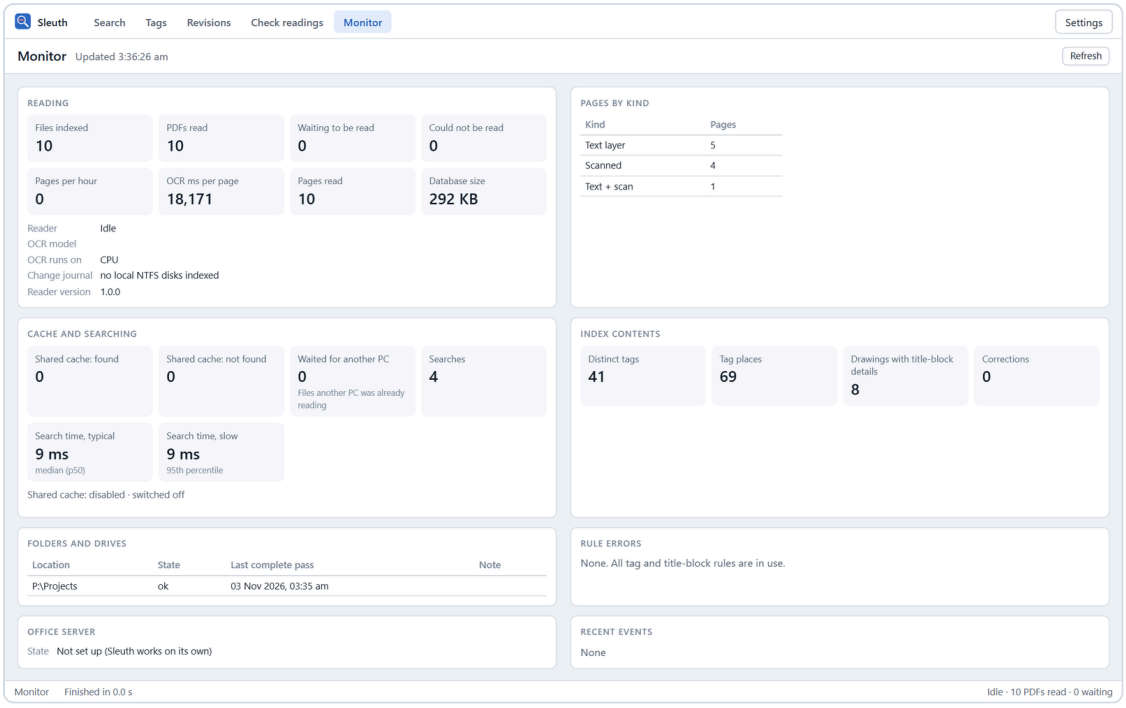
The **Check readings** view lists the lines OCR was least sure of, worst first, with the image of each beside it.

1. Look at the image and the reading. If the second reader (Windows OCR) read it differently, both readings are shown.
2. If the reading is wrong, type the text exactly as printed and click **Save correction**.
3. Searches and the Tags view now find your reading as **checked**. **Undo** removes it.

> **Nothing is learned**
> The original OCR text is kept beside your correction, and the OCR models never change. A correction applies to that file only, as long as its content is the same; if the drawing is replaced, the correction is kept for the record but no longer applied.

14 Monitor

The **Monitor** view shows how reading is going and why: useful when something seems slow or missing.

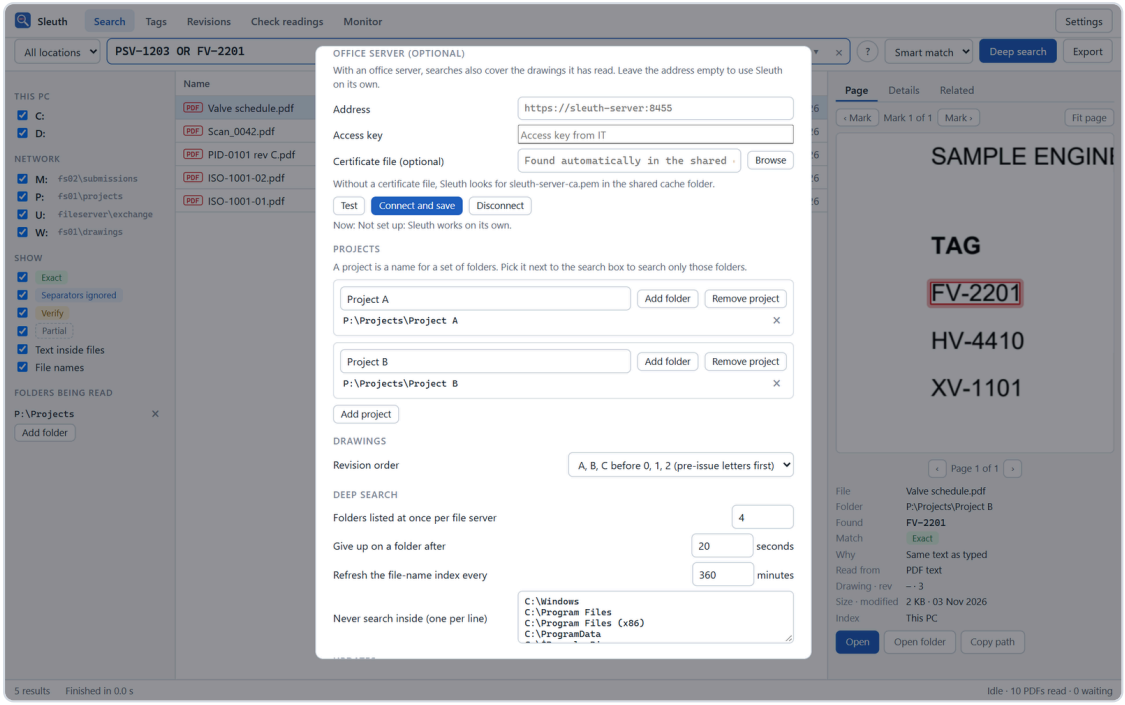


Everything the reader is doing, with the reasons.

- **Reading**: files read, waiting and failed; pages per hour; OCR time per page.
- **OCR runs on**: the CPU, or a graphics card through DirectML. Sleuth uses a graphics card only after it has read a check picture exactly like the CPU and was clearly faster; the line below says why it chose what it chose. A card that fails is not used again.
- **Cache and searching**: drawings taken from the shared cache, files another PC was already reading (Sleuth waits for its result instead of reading the same file), and search times.
- **Folders and drives**: the last complete pass of each location, and any that were offline or only partly listed.
- **Rule errors**: tag or title-block rules that could not be used.

15 Projects and the office server

Both are optional. Sleuth works fully on its own.



Settings → Office server and Projects.

Projects

A project is a name for a set of folders. Add them in Settings → Projects, then pick the project next to the search box to search only its folders. The Tags and Revisions views have the same project filter.

The office server

If IT runs the Sleuth office server, it reads the shared project folders once for everyone. Your searches then include its results (marked **Office**), and your PC leaves those folders to the server.

1. Get the server address and your personal access key from IT.
2. Settings → Office server: enter both. The certificate is found automatically in the shared cache folder; IT may give you a file instead.
3. Click **Test**, then **Connect and save**. Your key is stored encrypted for your Windows account.

You only see projects IT has given you. If the server is offline, Sleuth says so above the results and keeps searching your own index. **Disconnect** returns to working on your own.

16 The shared reading cache

Reading a scanned sheet takes about a minute. The shared cache means that minute is spent once for the whole office, not once per PC.

When any PC finishes reading a PDF on a network drive, it saves the result in the shared cache folder (by default `U:\Sleuth-cache`). Before reading a PDF, every other PC checks the cache first. If the same drawing is there, it is taken from the cache in an instant. Files are matched by their content, so a copied or renamed drawing is recognised too.

Why Sleuth insists on it

Without the cache, every PC would read every drawing itself: slower for you, and much more load on the network. So the cache is required. If it isn't working, Sleuth:

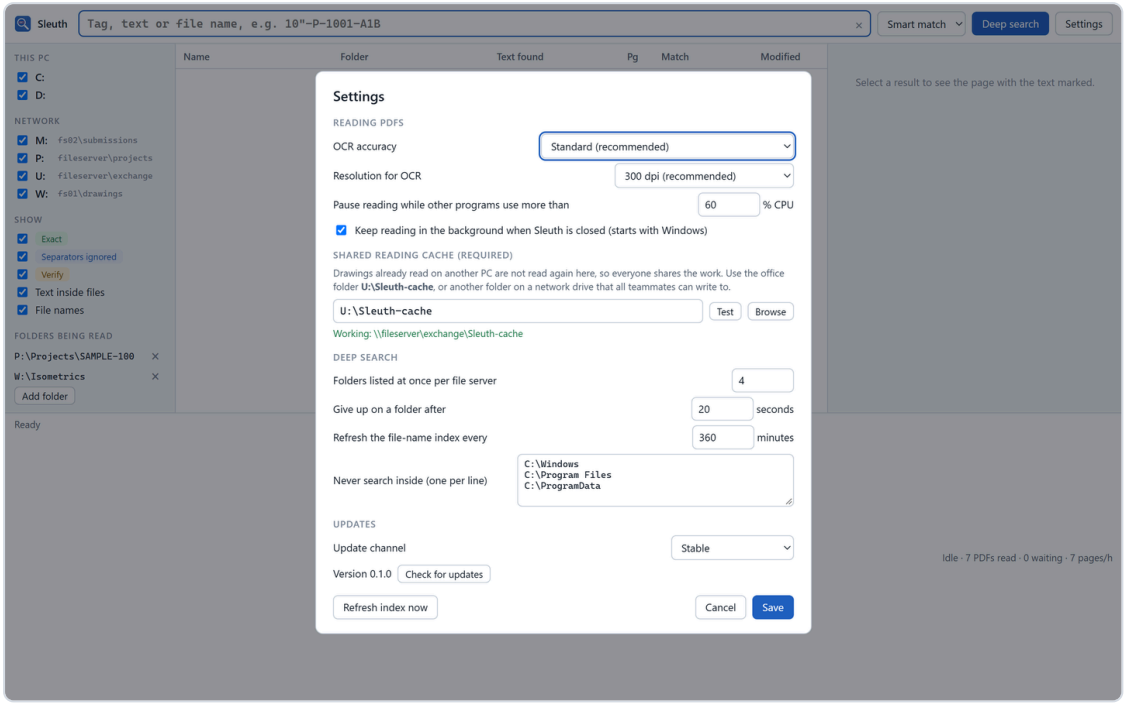
1. shows a yellow message with **Set it up now**;
2. after a minute, opens Settings with the cache section highlighted;
3. won't save a cache folder that doesn't work.

If the cache is only temporarily unreachable (you're off-site, or the server is down), Sleuth shows a quiet notice instead and keeps reading on your PC. It starts using the cache again by itself when it's back.

> **Don't delete files in the cache folder**
> They're the office's shared reading results. Deleting them only makes PCs read those drawings again.

17 Settings reference

Open with **Settings** at the top right.



The Settings window.

Setting	What it does	Default
OCR accuracy	**Standard** is recommended. **High** uses a larger model, downloaded once (140 MB); it is about 3× slower and found no extra text in our tests.	Standard
Resolution for OCR	300 dpi reads 2.5 mm text on A1/A3 sheets reliably. Use 400 dpi only for very small print; it is slower.	300 dpi
Pause reading while other programs use more than …	Reading waits while your PC is busy with other work.	60% CPU
Keep reading in the background	The reader keeps working when the window is closed, and starts with Windows.	On
Shared reading cache	The office folder for shared results. Required; see section 16.	`U:\Sleuth-cache`
Folders listed at once per file server	How hard deep search works each server. Raise for speed, lower if a server struggles.	4
Give up on a folder after	A folder that doesn't answer in this time is skipped and listed.	20 s
Refresh the file-name index every	How often the list of file names is rebuilt.	360 min

Never search inside	Folders Sleuth never walks or reads, one per line.	Windows and program folders
Update channel	**Stable** for everyone. **Development builds** only if you're testing new versions.	Stable
Refresh index now	Rebuilds the file-name list straight away, in the background.	

18 Updates

Sleuth keeps itself up to date. Your settings and index are always kept.

- Sleuth checks for a new version when it starts and every 6 hours. When there is one, a blue bar appears at the top: **Restart and update**.
- Sleuth downloads the new installer, checks it against the published checksum, stops the reader, installs, and starts again, usually in under a minute.
- To check yourself: **Settings** → **Check for updates**.
- The Windows "protected your PC" prompt does not appear for in-app updates.

Rolling back

If a new version causes trouble, **Settings** → **Updates** → **Roll back to version X** reinstalls the release before it, checked the same way. Settings and the index are kept; an index made by 1.0 can still be opened by 0.1, which simply does not show the 1.0 views.

> **Never lose your setup**
> Updates only replace program files. `settings.json`, your folders being read, and the index stay exactly as they were.

19 Troubleshooting and FAQ

Problem	What to do
A drawing I know has the tag isn't found	Is its folder in **Folders being read**? If not, add it, or right-click the file and choose **Read this PDF now**. If you don't know where it is, use **Deep search**. Also try **Smart match** rather than Exact only.
Results say "Verify, OCR unsure"	Open the preview and read the tag yourself. Faint, skewed or handwritten text is harder to read; the label tells you to look.
"The shared reading cache isn't working"	Map `U:` or choose the folder IT gave you, in Settings → Shared reading cache → **Test** → **Save**.
Reading seems slow	Scans take about a minute per sheet. Reading pauses while you're busy and runs faster when you step away. Check the status bar: "paused: PC busy" is normal.
A drive shows "Offline, skipped"	The share didn't answer. Check it in Explorer; when it's back, search again.
"N folders not searched"	Those folders were locked or too slow. Hover to see them. Raise "Give up on a folder after" if a server is just slow.
The window says the reader is restarting	Wait a few seconds; Sleuth restarts the reader automatically.
Windows blocked the installer	Click **More info** → **Run anyway**. This happens only for the first install.
I want to start the index from scratch	Close Sleuth, delete `%LOCALAPPDATA%\Sleuth\index.db`, and open Sleuth again. Results from the shared cache come back quickly.

Questions people ask

Does Sleuth send my drawings anywhere? No. Everything is read on your PC. Results are stored on your PC and in the office's shared cache folder. Nothing goes to the internet; the only internet traffic is checking GitHub for updates.

Does it change my files? Never. Sleuth only reads files.

Can it search DWG or Word files? Version 0.1 searches inside PDFs, and finds any file type by name.

Why does a file appear once even if the tag is on five pages? One row per file keeps the list readable. The preview steps through every mark and page.

20 Your data, privacy and uninstalling

- **On your PC**: settings in `%APPDATA%\Sleuth`; the index, models and logs in `%LOCALAPPDATA%\Sleuth`.
- **On the network**: read results in the shared cache folder, readable by the office.
- **On the internet**: nothing, except update checks to GitHub.
- **Logs**: `%LOCALAPPDATA%\Sleuth\logs\worker.log` is useful to IT when something goes wrong.

Uninstalling

Settings → Apps → **Sleuth** → Uninstall. This removes the program, Start-menu entry and start-up entry. Your settings and index are left in place in case you reinstall; delete the two folders above to remove them too.

21 Quick reference card

Print this page and keep it by your desk.

I want to…	Do this
Find a tag	Type it. Dashes, quotes and spaces are optional.
See where it is on the sheet	Click the result; the red circle marks it. **Mark** › for the next one.
Open the drawing	Double-click, or `Enter`.
Open its folder	Right-click → **Open folder**.
Send a colleague the location	Right-click → **Copy network path**.
Search everywhere, live	**Deep search**, or `Ctrl` + `Enter`.
Make a folder searchable inside	**Folders being read** → **Add folder**.
Read one file right now	Right-click → **Read this PDF now**.
Check what's been read	Click the status at the bottom right.
Find any of several tags	`PSV-1203 OR PSV-1204`
Only this project	Pick it next to the search box.
See every place a tag appears	**Tags** view → click the tag.
Compare two revisions	**Revisions** view → expand the drawing number.
Fix a wrong OCR reading	**Check readings** → type the text → **Save correction**.
Fix the shared cache	Settings → Shared reading cache → **Test** → **Save**.

Label	Trust it?
Exact	Yes.
Separators ignored	Usually; glance at the preview.
Verify, OCR unsure	Check the page yourself before relying on it.

Sleuth 0.1 · Dexterity Design Services Pvt. Ltd. · Screenshots show Sleuth's own interface with results read from sample drawings; folder and server names are examples. Latest version and this guide: github.com/dexterity-admin/Sleuth